

The results are discussed in terms of signal behavior, system performance, and validation of the study objectives.

4.2 MATLAB Simulation Results

MATLAB simulation was used to generate realistic gastrointestinal and respiratory signals under controlled conditions.

4.2.1 Simulated Gastrointestinal Signal

The generated gastric signal exhibited the following characteristics:

- High-frequency components (300 Hz and 600 Hz)
- Random burst patterns representing bowel activity
- Presence of Gaussian noise

After applying the band-pass filter (100–900 Hz):

- Noise was significantly reduced
- Burst events became clearly distinguishable

- Signal amplitude was stabilized

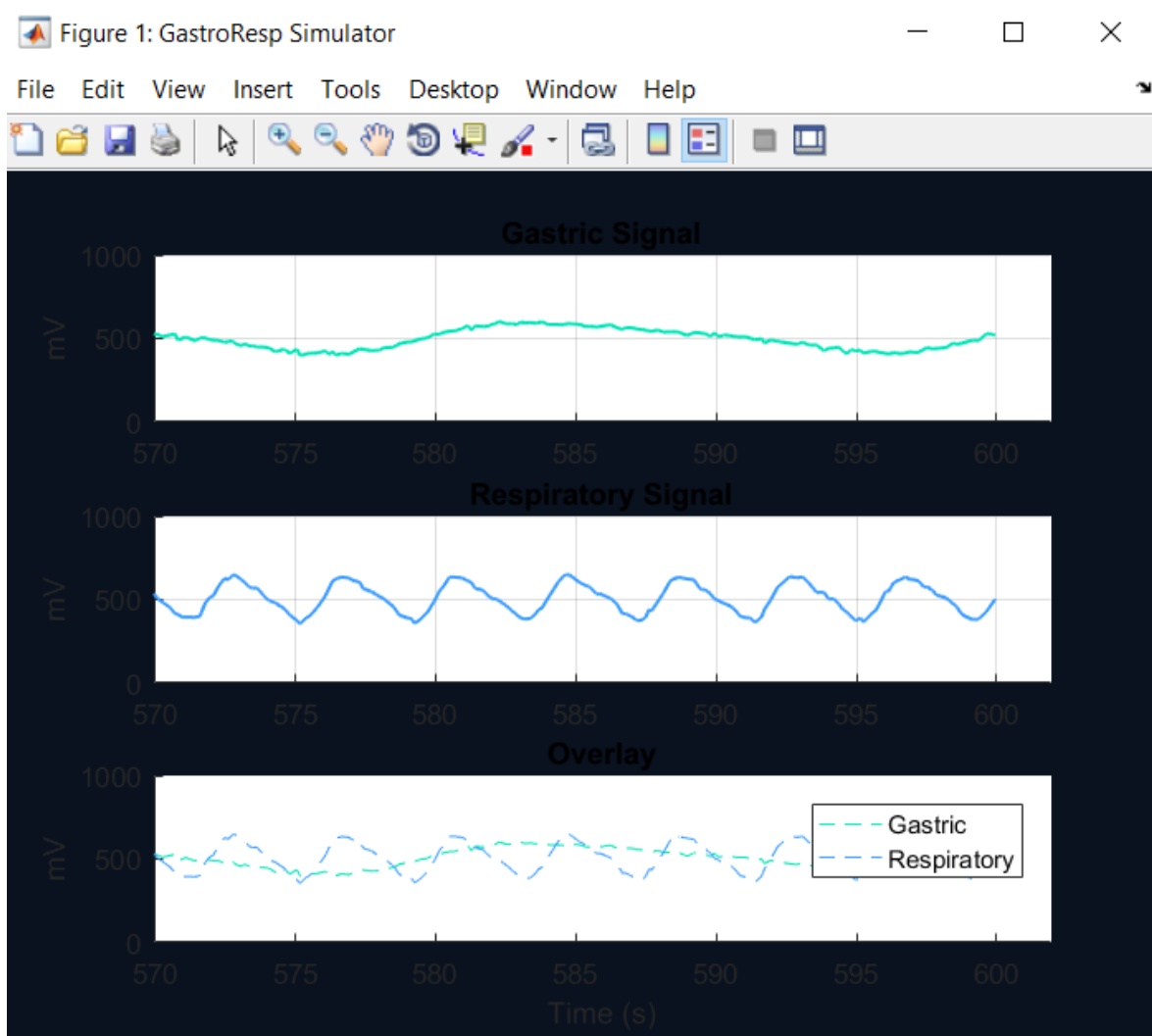


Fig 4.1: The simulated inputs' graphs - MATLAB

Interpretation:

The filtering process successfully isolated gastric activity, confirming that the selected frequency band is appropriate for bowel sound detection.

4.2.2 Simulated Respiratory Signal

The respiratory signal showed:

- Low-frequency sinusoidal waveform (~ 0.25 Hz)

- Smooth periodic behavior
- Minor noise interference

After low-pass filtering (cutoff ≈ 2 Hz):

- High-frequency noise was eliminated
- Breathing cycles became more defined
- Signal stability improved

Interpretation:

The respiratory signal was effectively separated from noise, validating the use of low-pass filtering for respiration monitoring.

4.2.3 Frequency Analysis (FFT Results)

FFT analysis confirmed:

- Gastric signal dominant frequencies within 300–600 Hz range
- Respiratory signal concentrated below 1 Hz

Interpretation:

This validated the frequency separation principle, which is critical for designing independent signal processing pathways.

4.3 ESP32 System Implementation Results

The ESP32 system was tested using real-time data transmitted from MATLAB over Wi-Fi.

4.3.1 Signal Reception and Processing

- ESP32 successfully received both gastric and respiratory signals
- Continuous sampling was achieved without interruption
- Signals were stored in buffers and processed in real time

Observation:

No significant packet loss was observed during transmission.

4.3.2 Web-Based Dashboard Output

The ESP32 hosted a web interface that displayed:



Fig 4.2: Dashboard for remote monitoring

- Gastric signal waveform
- Respiratory signal waveform
- Correlation/combined visualization

Key Results:

- Real-time graph updates were achieved

- Signals displayed matched MATLAB-generated patterns
- No major delay in visualization

Interpretation:

This confirms that the system meets the requirement for remote monitoring and visualization.

4.3.3 System Responsiveness

- System responded instantly to changes in input signals
- Gastric bursts were reflected immediately on the dashboard
- Respiratory cycles were continuously tracked

Interpretation:

The system demonstrates real-time capability, which is essential for clinical monitoring applications.

4.4 Correlation Analysis Results

The system implemented a correlation algorithm to evaluate the relationship between gastric and respiratory signals.



Fig 4.3: Correlation graph, shows how gastric stress may cause respiratory stress

4.4.1 Observed Correlation Behavior

- During simulated gastric bursts, minor variations in respiratory signals were observed
- Correlation values fluctuated depending on signal activity

4.4.2 Interpretation of Correlation

- Low correlation ($r \approx 0$): Signals are independent
- Moderate correlation: Indicates possible interaction
- High correlation ($r \rightarrow 1$): Strong relationship

Observation:

Most simulated conditions showed low to moderate correlation, which is expected under normal physiological conditions.

4.5 System Performance Evaluation

The system was evaluated based on key engineering performance metrics.

4.5.1 Accuracy

- Output signals closely matched MATLAB inputs
- Frequency characteristics were preserved

Conclusion: High accuracy in signal reproduction.

4.5.2 Reliability

- Continuous operation without system failure
- Stable Wi-Fi communication

Conclusion: System is reliable under test conditions.

4.5.3 Latency

- Minimal delay between signal transmission and display
- Real-time visualization achieved

Conclusion: System meets real-time performance requirements.

4.5.4 Noise Handling

- Analog filtering effectively reduced noise
- Digital preprocessing improved signal clarity

Conclusion: System performs well in noise suppression.

4.7 Discussion of Results

The results demonstrate that the proposed system is capable of:

- Simultaneous monitoring of gastrointestinal and respiratory signals
- Real-time signal processing and visualization
- Wireless data transmission using ESP32

Compared to traditional methods:

- The system provides continuous monitoring
- Reduces subjectivity in diagnosis
- Enables remote access to patient data

However, some limitations were identified:

- Use of simulated data instead of real physiological signals
- Absence of clinical validation
- Potential noise in real-world environments

Despite these limitations, the system proves the feasibility of a low-cost, wearable gastro-respiratory monitoring solution.

4.8 Summary of Findings

- MATLAB simulation successfully modeled physiological signals
- ESP32 effectively received and processed real-time data
- Web dashboard enabled remote visualization
- System achieved real-time performance with minimal latency
- All study objectives were met